# SoK: Reusing Defense Evidence in LLM and Agent Side Channels

***Abstract*—Side-channel defenses for LLM serving and agents intervene in caches, traffic and tool actions. This SoK maps 82 study families, including 52 core privacy or attribution families; 19 have an evaluated mitigation record under our selection criteria. We systematize those interventions by mechanism and generation layer, and distinguish their measured outcomes from formal guarantees and unresolved residual channels. Across the 19 families, an inference outcome is explicitly bound in 15 of 16 applicable families, utility in 3 of 16, and a matched clean/protected fitting contrast in 2 of 10. Accepting partial evidence raises utility to 5 of 16 while the fitting contrast remains 2 of 10. Five of eight families with explicit inference and cost place them in different records. Agent privacy and attribution studies provide partial usefulness evidence but no explicit task-utility binding in the four-family subset. Source-linked examples connect these patterns to a deployment choice: retain a measured configuration, separate a cost from a different primitive, or measure state and task outcomes at the chosen execution boundary.**

## 1. Introduction

Encryption conceals message content, yet LLM serving and agents expose packet sizes, cache timing, destinations and tool sequences [1], [2], [3], [4], [5], [6]. Padding, cache isolation and decoy actions address different parts of this problem. Choosing among them requires both a protection result and evidence of what the protected system still does, at what cost.

What the literature reports together. In the inspected privacy and attribution mitigation sources, inference is explicitly bound in 15/16 applicable families, utility in 3/16 and a matched clean/protected fitting contrast in 2/10. Accepting partial evidence gives 16/16, 5/16 and 2/10. Cost is explicit in 11/18 families, yet five of eight families with both inference and cost place them in different records. This reporting pattern motivates a deployment question: *which evidence describes the chosen intervention, and what additional measurement does a change of receiver or execution require?*

One example, three decisions. Consider a serving operator choosing a cache-reuse defense. Early Bird evaluates a prefix-recovery oracle and separately measures semantic-cache deidentification overhead [3, V-A–B]. CachePrune evaluates direct and contextual recovery and links its default configuration to generation quality and serving cost [7, 5.1–5.6]. Prefix and direct-recovery results inform a shortlist for a shared token-recovery threat. Early Bird's semantic-cache overhead prices a different mechanism, whereas CachePrune's shared package informs feasibility at its tested configuration. A changed workload then identifies the next workload and cost measurements. This example connects the reporting pattern to a concrete selection decision.

The 82-family map supplies context. The privacy and attribution synthesis covers 31 records from 19 families; two evasion records and six contextual records retain comparison boundaries. We classify mitigations by mechanism and generation layer, and changes by *observed signal*, *executed intervention* and *evaluation setting*. A cover action can change the environment, and component output quality can differ from task success.

Contributions.

- **A defense landscape and reporting synthesis.** A mechanism-by-layer map locates evaluated interventions and unmapped privacy/attribution families. Source-linked censuses expose inference, utility, adaptation and cost reporting, with status and denominator sensitivities (§§4–5).
- **Configuration-level comparison and positive reuse.** The record atlas separates costs measured for another mechanism and retains outcomes measured together, including the CachePrune package and SparLeak's intensity-specific inference/quality pairs. Exact record identities make each combination reconstructable in this PDF (§5).
- **Deployment measurement plans.** The holdout synthesis separates attack generalization from tested defense transfer. A reusable reporting card and filled examples translate a changed signal, intervention or setting into retained evidence and a specific next measurement (§§6–8.1).

The synthesis builds on website-fingerprinting and adaptive-attack evaluation practice [8], [9], [10], [11]. It connects that practice to the measured objects in LLM/agent systems, where serving primitives, tool state and planner utility coexist. Worked applications illustrate the resulting decisions.

**Reading the findings.** Counts describe bindings located in the inspected sources, under the stated applicability rules. An unreported binding is an open measurement for the proposed reuse; it does not establish that the source authors never performed the experiment.

TABLE 1. RESEARCH QUESTIONS, SOURCE-BASED FINDINGS AND DEPLOYMENT CONSEQUENCES.

| Question | Synthesis finding | Concrete use / evidential basis |
|---|---|---|
| RQ1: landscape | Evaluated mitigation covers 10/22 pure service-layer families, 1/8 pure hardware and 2/13 attribution families. | Target underrepresented settings within the empirical core map; preserve its C1–C5 eligibility. |
| RQ2: comparability | Five of eight families with explicit inference and cost place them in different records. | Keep Early Bird prefix recovery on the shortlist while obtaining a cost for that primitive. |
| RQ3: joint evidence | Explicit inference 15/16; utility 3/16; matched adaptation 2/10. With partial evidence: 16/16, 5/16, 2/10. | Request a task outcome for agent privacy cover; retain configuration-linked packages such as CachePrune. |
| RQ4: transfer | Three model and two topology holdouts concern attack generalization. | Reuse the tested attacker coverage; separately establish source-to-target defense-calibration outcomes. |

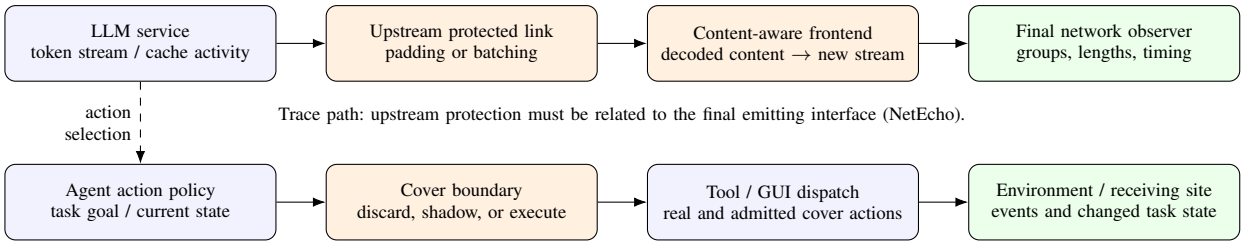



Figure 1. Receiving views and execution boundaries. A content-aware frontend can emit a new trace; admitted cover actions can change state. Evidence must describe the object a deployment will expose or execute.

## 2. Scope and Systematization Method

### 2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions. **RQ1 (landscape):** which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems, and where is each signal generated versus observed? **RQ2 (comparability):** when do two reported defense results measure the same target, receiver, intervention and cost object? **RQ3 (joint evidence):** which results bind protection, utility and cost at one evaluated operating point? **RQ4 (transfer):** which evidence distinguishes attack generalization, defended-condition evaluation and defense transfer across tasks, models, topologies and hardware? RQ1 supplies the field context; RQ2–RQ4 correspond to the three contributions.

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [12], [13], [14], [15], [16]. Table 2 states the observable eligibility rules. For C3, the extraction must name an execution mechanism and an evaluated observable separate from ordinary linguistic meaning. A filter qualifies when its accept/reject behavior or differential response is the measured oracle for a hidden property; merely analyzing a reply's wording or style does not. If a mixed paper does not isolate that mechanism, its membership stays adjacent. This operational distinction concerns the measured signal, not whether the output is text.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

| Observed input | Decision and required distinction |
|---|---|
| Length, timing, cache response | Core when measured execution dependence supports hidden-target inference. |
| Own-site actions | Core attribution with site permission; no automatic access to other sites or private tool logs. |
| Reply states the secret | Direct disclosure; adjacent, not auxiliary-channel evidence. |
| Reply style/content identifies a user | Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator. |
| Output-filter membership oracle | Core only for an isolated accept/reject or differential-response oracle tied to filter execution; semantic content alone remains adjacent. |
| Privileged loop/fault logs | Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence. |

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns

evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [17], [18]. These roles are mapped per core family, alongside generating and receiving layers, in Table 7.

### 2.2. Corpus construction and reconciliation

Forty seeds, venue-targeted proceedings screening and one-hop citation snowballing produced an initial map of 60 families (43 core, 17 adjacent). A supplementary database and bulk keyword search returned 615 occurrences and 565 unique records; auditing its 204 exclusions rerouted four records, and the search added 15 core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The completed search frames, closed on October 21, 2026, combine Semantic Scholar and native arXiv queries with venue-targeted screening of S&P, USENIX Security, NDSS, CCS and PMLR. C1–C5 screening and version deduplication operate on records before family reconciliation; the supplement retains queries, dispositions and excluded incomplete search frames. Architecture and hardware venues were not comprehensively screened; the finite frame is not an exhaustive census.

### 2.3. Source-bound result records

Each record links its target, observation boundary and operating point to a source version and section/table/figure locator. The authors report reviewing the classification of all 82 families using titles, abstracts and experimental sections, and checking all 39 result records and 180 family-audit cells against source locators. The subsequent 351-cell record audit and 133 within-question pair decisions retain author-assigned reasons; executable checks establish their accounting consistency. Candidate fields produced with AI assistance were subject to this author review.

Appendices B and F contain the record index, comparison summary and query memberships in this PDF. The ledger also separates cited publication status, inspected version, measurement mode and remaining binding uncertainty. Published status increases traceability of venue review but does not establish the correctness of a result; a preprint's observations remain provisional at their inspected scope.

**SoK Takeaway 1 (Map evidence to the receiving deployment).** A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

## 3. Result Records and Comparability Rules

The atomic unit of our systematization is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration ($D_0$: clean-condition fit; $D_2$: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A record is a comparison unit with a source-indexed parameter grid, not an independent trial. We split when a change of primitive, execution mode, target/receiver contract or separately measured cost object would alter what a result can support. We retain models, datasets, parameter values and repetitions as coordinates within that unit; numerical joins must preserve those coordinates. Source versions are reconciled within a family before extraction.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

| Record field | Required meaning |
|---|---|
| Source | Family/result ID, version, source locator, reading scope and hash. |
| Target | Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis. |
| Receiver | Actual permission and released representation, including session/document linkage and any auxiliary inputs. |
| Intervention | Where protection ends; whether edits affect stored traces, transmission or executed actions. |
| Attacker | Fit/calibration condition, test population, training access and adaptation budget. |
| Workload | Task, system, topology/hardware and grouping/holdout. |
| Outcomes | Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states. |

Indexing schema and comparison contracts. The index names the hidden property, measured channel, observer permissions, intervention and evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

Record boundaries and coverage. The 39-record ledger retains 31 privacy/attribution mitigation records from 19 families, two Type III evasion records and six contextual controls. The primary defense synthesis uses the first 31; the evasion records illustrate a different protected objective. A defense discussion without a scoped evaluated result stays in the map. S34 publishes the full record scopes and split decisions. Three examples fix the granularity:

- **Split:** Early Bird's $K$-prefix recovery oracle and semantic-cache processing benchmark change both

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

| Requirement | Condition and resulting license |
|---|---|
| Descriptive comparison | Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost. |
| Paired effect | Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved. |
| Deployment transfer | Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered. |

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

| Join key / field | Entry |
|---|---|
| Family / result | Dangol workflow / `DANGOL-livebinding`. |
| Version / locator | arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [19]. |
| Target / view | Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation. |
| Intervention | Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures. |
| Workload / testbed | Workflow classes with client, relay, gateway and agent co-located on one loopback host. |
| Outcome / cost | Accuracy 0.30; about $13\times$ bytes and $11\times$ latency in that condition. |
| Unresolved fields | Objective terminal task utility $\mathsf{U}$; precise paired live-refit comparison ?. |
| Licensed judgment | These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter. |

primitive and measured object; their numbers cannot form one protection–cost point.

- **Retain coordinates:** SparLeak's model-by-replacement-intensity ASR/ROUGE grid remains one mitigation record. Each intensity's inference/quality pair must be retained together; selecting the lowest ASR and highest ROUGE from different rows is disallowed.
- **Split outcome, share package:** CachePrune's direct recovery and contextual guessing require two target contracts. They reference one shared default quality/cost package, counted once when joining configurations.

Question cohorts precede compatibility judgments. Records enter one declared comparison question by the intervention and receiving object, not by score or verdict: delivered traffic, service-cache reuse, local memory/resource behavior, behavioral humanization, generator fingerprinting or fixed instructions. Three contextual singletons concern access gates, contention and filter-created views. S34 lists

TABLE 6. TAXONOMY OF CHANGES RELEVANT TO EVIDENCE REUSE.

| Changed object | Typical transition | Evidence to carry or renew |
|---|---|---|
| Observed signal | Receiver, released representation, hidden target | Keep the target and permissions; measure the actual receiving trace. |
| Executed intervention | Prefix/semantic cache; transformed trace/live scheduler; read/write cover | Keep mechanism and state boundary; bind function and cost to the executed configuration. |
| Evaluation setting | Attacker fit, workload, model, hardware or network | Keep training/test provenance; establish the proposed holdout and relevant baseline. |

operational membership rules and all 39 assignments. When a mechanism spans several questions, retain the primary extracted question and document the alternative; a new cross-cohort question requires a new inventory. A shared cohort admits a pair for inspection; it does not make different targets or costs compatible. These retrospective rules define the comparison index. Appendix F varies record granularity and retains the original judgments. The reporting-pattern synthesis in Section 5 uses item availability and within-record bindings directly.

A taxonomy of reuse changes. The categories below describe what changes between a source result and a deployment question; several can apply together. The mechanism-by-layer matrix in Section 4 adds the defense design space. The nine audit items identify the evidence needed to evaluate a selected mechanism at its receiver and execution boundary.

From contracts to an audit. Section 5 operationalizes these contracts as nine reuse items with a shared evidence-status alphabet (Table 11) and applies them to every mitigation family (Table 12).

## 4. Corpus Map Across 82 Study Families

### 4.1. Generating layers, receiving projections and threat polarities

Table 7 maps all 58 core families by signal-generating layers, receiving projections and threat polarity. Multi-layer families retain every reported layer in their generating and observing sets.

Cross-layer synthesis. A remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate why receiving permissions and protected objectives must accompany the generating layer [6], [17], [36].

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4

TABLE 7. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

| Core family | $G \to O$ | Type |
|---|---|---|
| **hardware generation, P1 (8)** | | |
| Edge EM leakage [20] | $1 \to 1$ | I |
| Energon [21] | $1 \to 1$ | II |
| MoEcho [22] | $1 \to 1$ | I |
| I Know What You Said [23] | $1 \to 1$ | I |
| Behind Bars [24] | $1 \to 1$ | I |
| PAGE404 [25] | $1 \to 1$ | I |
| Spill the Beans [26] | $1 \to 1$ | I |
| ArchSnoop [27] | $1 \to 1$ | II |
| **hardware and service, P1/P2 (8)** | | |
| SparLeak [28] | $1,2 \to 1$ | I |
| Shape and Substance [29] | $1,2 \to 1$ | I |
| TDXRay [30] | $1,2 \to 1$ | I |
| SparSEEty [31] | $1,2 \to 1$ | I |
| OTRO [32] | $1,2 \to 1$ | I |
| Detokenization Leaks [33] | $1,2 \to 1$ | I |
| LLM-WFIN [34] | $1,2 \to 1,3$ | I |
| LLM-serving RCA [35] | $1,2 \to 2$ | IV |
| **service generation, P2 (13 of 22)** | | |
| SplitDecoy [36] | $2 \to 2$ | I |
| KVGov [37] | $2 \to 2,4$ | I |
| LLMs Have Rhythm [38] | $2 \to 3$ | II |
| Remote timing [39] | $2 \to 3$ | I |
| NetEcho [40] | $2 \to 3$ | I |
| Prompt Echo [41] | $2 \to 3$ | I |
| Speculative decoding [42] | $2 \to 3$ | I |
| Remote keylogging [1] | $2 \to 3$ | I |
| Whisper Leak [43] | $2 \to 3$ | I |
| Cache contention [44] | $2 \to 4$ | I |
| Routing metadata [45] | $2 \to 4$ | I |
| SafeKV [46] | $2 \to 4$ | I |
| ML privacy channels [47] | $2 \to 4$ | I |

| Core family | $G \to O$ | Type |
|---|---|---|
| **service generation, P2 (9)** | | |
| Early Bird [3] | $2 \to 4$ | I |
| Prompt-cache audit [48] | $2 \to 4$ | I |
| InputSnatch [49] | $2 \to 4$ | I |
| Leaky Language Models [50] | $2 \to 4$ | I |
| PromptPeek [2] | $2 \to 4$ | I |
| SpliceLeak [51] | $2 \to 4$ | I |
| Time Will Tell [52] | $2 \to 4$ | I |
| CachePrune [7] | $2 \to 4$ | I |
| MoE prompt stealing [53] | $2 \to 4$ | I |
| **cross-layer sources (5)** | | |
| ClawGuard [18] | $1,5 \to 1$ | IV |
| MIRAGE [54] | $2,5 \to 2$ | IV |
| Cloud traffic benchmark [55] | $2,5,8 \to 3$ | I, II |
| Web-agent unmasking [56] | $3,7 \to 3,7$ | II, III |
| LumiMAS [57] | $4,5 \to 4,6$ | IV |
| **orchestration/workflow sources (15)** | | |
| AgentPrint [4] | $5 \to 3$ | I, II |
| Cycle detection [58] | $5 \to 4,6$ | IV |
| SIGLEAK [59] | $5,6 \to 4,6$ | I |
| ProcGrep [60] | $5,6 \to 4,6$ | II |
| AgentProv [61] | $5,6 \to 6$ | II |
| Lidar [62] | $5,6 \to 6$ | II |
| Dangol workflow [19] | $5,6,8 \to 3,6,8$ | I |
| Campaign attribution [63] | $5,6,8 \to 6,8$ | II |
| WRA [6] | $5,7 \to 3$ | I |
| PersonaFingerprint [64] | $5,7 \to 3$ | I |
| Whose Agent? [65] | $5,7 \to 7$ | II |
| AgentTell [66] | $5,7 \to 7$ | I |
| UI Traces [5] | $5,7 \to 7$ | II |
| FP-Agent [67] | $5,7 \to 7$ | II |
| AHB [17] | $5,7 \to 7$ | III |

$G$ denotes signal-generation layers and $O$ the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

Mitigation coverage is uneven across the map. Table 8 organizes the 19 privacy/attribution mitigation families by mechanism and generating layer. Within the 52-family Type I/II union, evaluated mitigation is linked to 1/8 pure hardware families, 4/7 hardware/service, 10/22 service-only, 3/13 workflow and 1/2 other cross-layer families. Two of 13 attribution families have such a record. Appendix G lists the other 33 families. These are gaps in the inspected C1–C5 frame; defense-only work outside that frame is not counted. The receiving view and remaining channel determine which mechanism and additional measurement are relevant.

## 4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [68], [69], [70]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [71], [8], [72], [73]. Mathews et al.

TABLE 8. PRIVACY AND ATTRIBUTION INTERVENTIONS INDEXED BY MECHANISM AND SIGNAL-GENERATION LAYER. A CELL NAMES FAMILIES WITH A QUALIFYING MITIGATION RECORD IN THE INSPECTED CORPUS; MULTIPLE MECHANISMS MAY OCCUR IN ONE FAMILY. LETTERS DENOTE THE EVIDENCE PRESENT IN THE INDEXED RECORD, NOT A UNIVERSAL SECURITY GUARANTEE: E = EMPIRICAL PROTECTED-VIEW OR SIMULATED ATTACK OUTCOME; C = COMPONENT COST OR PROPOSAL/ESTIMATE WITHOUT A BOUND ATTACK OUTCOME IN THAT INDEXED RECORD. THE TABLE DOES NOT ADJUDICATE WHETHER EACH SOURCE ALSO PROVES A FORMAL GUARANTEE.

| Mechanism | P1 only | P1/P2 | P2 only | Workflow | Other cross-layer |
|---|---|---|---|---|---|
| Traffic framing / rate | – | – | NetEcho, Whisper, SPEC, Remote, Time [40], [43], [42], [39], [52] (E/C) | Dangol [19] (E) | – |
| Cache isolation, policy or timing | – | – | Early Bird, SpliceLeak, SafeKV, KV-Gov, CachePrune [3], [51], [46], [37], [7] (E/C) | – | – |
| Memory access / isolation | Gao [23] (C) | TDXRay, OTRO [30], [32] (C/E) | – | – | – |
| Model computation or output shaping | – | SparLeak, Shape [28], [29] (E/C) | – | – | – |
| Decoys, timing edits, or input changes | – | – | – | WRA, UI Traces [6], [5] (E) | Cloud benchmark [55] (E) |

Cells concern the mapped generating layer, not necessarily the intervention layer. Residual observations are receiver-specific: grouped or re-emitted traffic can remain after framing; timing can remain after a cache-policy change; access counts can remain after address hiding; and action types/order can remain after timestamp edits. A dash means no matching indexed record in this collection, not that the mechanism cannot apply or that no external defense exists. Type III evasion and Type IV authorized diagnostics are boundary cases, not privacy-protection results.

re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [9]. Adaptive attacks and online-cost evaluation are established foundations.

Adaptive evaluation and benchmarking. Tramèr et al. show why defense-specific adaptive attacks require more than applying an unchanged attack; Carlini et al. set out robustness-evaluation practices; RobustBench standardizes the evaluated threat and test interface [10], [11], [74]. These works already motivate protected-data evaluation and reproducible benchmarks. Our R4/R5 items instantiate that discipline. The LLM/agent-specific question is whether a refit, cost or functional result belongs to the same prefix policy, content-aware emitter or state-changing action under discussion.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record atlas documents these distinct measurement objects; the 82-family holdout matrix separates evidence for each transfer axis. This connects evidence that earlier taxonomies organize by attack or mechanism: one family can contain several execution modes, a receiving projection can cross system layers, and utility/cost can belong to a different operating point from inference. The synthesis preserves these distinctions when translating a reported result into a deployment decision.

Formal guarantees and runtime measurements. The mitigation inventory covers evaluated runtime results, not the complete literature on privacy proofs. Differential privacy constrains releases for a specified adjacency relation and privacy budget [75]; information-flow measures distinguish mutual information from single-guess vulnerability [76]. Constant-time verification and oblivious memory protect stated traces under their own leakage models [77], [78]. These guarantees can supply stronger source-scoped evidence than a low empirical attack score. Their scope still matters: a guarantee for model weights need not include a serving trace, and hiding addresses need not hide access counts. TDXRay and OTRO connect these foundations to the indexed LLM components. For reuse, retain the stated adjacency, leakage model and execution assumptions. Attack-refitting items are inapplicable to a purely formal proof; absence of a classifier experiment does not count against that guarantee.

## 5. Reporting Patterns and Joint Defense Evidence

### 5.1. Evidence frame and reporting rules

The 58 core families contain 52 with a privacy or attribution objective, including 19 with a scoped mitigation record. We organize the 31 records of those 19 families by mechanism, target layer, receiving view and measured outcome (Tables 8 and 9). The archival 39-record ledger also contains two AHB evasion records and six contextual controls; these eight do not enter the privacy/attribution denominators. Type IV operator diagnostics remain in the landscape map as a distinct authorization boundary.

Table 11 defines the nine evidence items for a requested reuse. Component quality qualifies as utility for a component claim; an adaptation comparison requires the corresponding attacker-fit evidence. We derive each family cell from the record ledger, retaining B if any record supplies it, otherwise P, ?, U, A, then inapplicable. This availability projection keeps the record identities and source locators. Different columns can refer to different configurations, so configuration-level combinations are checked within a record.

TABLE 9. SEVEN PRIVACY AND ATTRIBUTION INTERVENTION CATEGORIES ORGANIZED BY RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. TYPE I PROTECTS CONFIDENTIALITY; TYPE II ADDRESSES PROVENANCE OR IDENTITY ATTRIBUTION.

| Intervention & polarity | Evaluated view, target and source | Established result and applicable boundary |
|---|---|---|
| Padding / batching (Type I: confidentiality) | Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [40], [43]. | Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U. |
| Constant-rate / transport cover (Type I: confidentiality) | Network workflow fingerprint; Dangol's OHTTP testbed [19]. | The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility. |
| Concurrent query decoys (Type I: confidentiality) | Destination/domain observer; private prompt and trait inference. WRA [6]. | Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy $E_{\text{func}}$ from 0.770 to 0.610. Terminal decoy utility/cost are U. |
| Cache isolation / keying (Type I: confidentiality) | Co-tenant prefix timing; SafeKV, KVGov and CachePrune [46], [37], [7]. | SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median $1.3\,\mu$s prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance. |
| Cache timing equalization (Type I: confidentiality) | Co-tenant cache-fusion timing; SpliceDefense [51]. | Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and $<$ 5ms/$<1\%$ prefill time; this does not price combined deployment. |
| Timestamp perturbation (Type II: attribution) | Recorded website action metadata; model/agent attribution. UI Traces [5]. | Fitting on delayed traces ($D_2$) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation. |
| Oblivious / unshared memory (Type I: confidentiality) | Host or local resource observations; TDXRay, Gao et al. and OTRO [30], [23], [32]. | TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds. |

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

TABLE 10. EVIDENCE REPORTING IN THE 19 PRIVACY/ATTRIBUTION MITIGATION FAMILIES. B DENOTES EXPLICIT SOURCE-SCOPED BINDING; P DENOTES PARTIAL BINDING. DENOMINATORS DIFFER BY ITEM APPLICABILITY.

| Reported item | B / applicable | B+P / applicable |
|---|---|---|
| Inference outcome (R7) | 15/16 | 16/16 |
| Cost (R9) | 11/18 | 15/18 |
| Utility (R8) | 3/16 | 5/16 |
| Protected fitting (R4) | 4/10 | 8/10 |
| Matched adaptation (R5) | 2/10 | 2/10 |

## 5.2. Inference is reported more consistently than utility or matched adaptation

Table 10 puts the reporting asymmetry first. Inference is explicitly bound in 15/16 applicable families, compared with 3/16 for utility and 2/10 for a matched clean/protected-fitting contrast on defended tests. Cost is bound in 11/18. Restricting to the same 13 families eligible for inference, utility and cost gives 12, two and six explicit bindings, respectively.

Sensitivity distinguishes two adaptation questions. Accepting P raises inference to 16/16, utility to 5/16 and cost to 15/18, while matched adaptation remains 2/10. Protected fitting rises from 4/10 to 8/10, distinguishing a partial description of fitting from an explicitly matched contrast. In the common 13-family population, accepting P gives 13 inference, four utility and ten cost bindings. Treating all 19 families as applicable for the sensitivity check gives 4/19 protected fitting and 2/19 matched adaptation, or 8/19 and 2/19 with P. Removal checks and member identities are in S41.

## 5.3. Agent privacy evidence lacks an explicit task-utility binding

We select agent privacy/attribution families by P5 generation and Type I or II in the core map. All four members are visible: UI Traces (R8 U), WRA (P), Dangol (U), and the cloud traffic benchmark (P). WRA reports separately judged usefulness; the cloud benchmark asserts semantic preservation. Neither provides the same-condition task score required by R8 B. Hence the subset has 0/4 explicit and 2/4 explicit-or-partial utility bindings. This pattern survives both removal checks. It is a gap in *explicit task-utility binding*, not absence of all functional evidence.

The three Type I/II R8 B families show what existing utility evidence can support. TDXRay binds vocabulary coverage to lookup behavior; CachePrune binds generation quality to its serving configuration; SparLeak pairs recovery and ROUGE by intensity [30], [7], [28]. AHB binds task accuracy for Type III evasion and is retained only as a boundary example [17]. This distinction identifies the task metric needed when a privacy cover method is moved into an acting planner.

## 5.4. Family-level inference and cost often describe different records

Eight families have explicit inference and explicit cost somewhere in their evaluations. In five—Early Bird, Splice-

TABLE 11. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

| Item | Reuse query → required binding (**B**) |
|---|---|
| R1 Target | Same protected interest → explicit hidden property, inference goal and authorization relation. |
| R2 Receiver | Same observer protection → permissions, released view and auxiliary inputs. |
| R3 Intervention | Same executed mechanism → placement/mode: stored data, wire emission or actions. |
| R4 Protected fit | Resistance to protected-data fitting → identified protected training/calibration and testing. |
| R5 Matched adaptation | Effect of adaptation → clean/protected fits tested on the same defended population. |
| R6 Workload | Changed workload/system → tasks, configuration and the relevant holdout axis. |
| R7 Inference | Quantified protection → metric, denominator and scoped defended evaluation condition. |
| R8 Utility | Preserved task/component function → performance at the same protected state/operating point. |
| R9 Cost | Resource-budget feasibility → units, baseline and cost object at that operating point. |

**B**: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. For R4/R5, – means that a training-fit contrast is inapplicable; it does not establish that non-learning adaptive attacks are irrelevant or absent. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

TABLE 12. EVIDENCE AVAILABILITY IN THE 19 PRIVACY/ATTRIBUTION MITIGATION FAMILIES. ROWS ARE RECORD-DERIVED PROJECTIONS; COLUMNS MAY REFER TO DIFFERENT CONFIGURATIONS.

| Family | R1 | R2 | R3 | R4 | R5 | R6 | R7 | R8 | R9 |
|---|---|---|---|---|---|---|---|---|---|
| UI Traces [5] | **B** | **P** | **B** | **B** | **B** | **B** | **B** | U | – |
| WRA [6] | **B** | **P** | **B** | – | – | **P** | **B** | **P** | U |
| NetEcho [40] | **B** | **B** | **B** | **B** | ? | **B** | **B** | U | **P** |
| Early Bird [3] | **B** | **B** | **B** | **B** | ? | **B** | **B** | U | **B** |
| Whisper Leak [43] | **B** | **B** | **B** | **P** | U | **B** | **B** | U | **P** |
| SpliceLeak [51] | **B** | **B** | **B** | – | – | **B** | **B** | U | **B** |
| Time Will Tell [52] | **B** | **B** | **B** | **P** | U | **B** | **B** | U | **B** |
| I Know What You Said [23] | **B** | **B** | **B** | – | – | **P** | – | – | **B** |
| Speculative decoding [42] | **B** | **B** | **P** | **P** | U | **B** | **B** | – | **B** |
| Dangol workflow [19] | **B** | **B** | **B** | **B** | **B** | **B** | **B** | U | **B** |
| Cloud traffic benchmark [55] | **B** | **P** | **B** | U | U | **P** | **B** | **P** | U |
| Remote timing [39] | **B** | **B** | **P** | – | – | **B** | – | U | **B** |
| TDXRay [30] | **B** | **B** | **B** | – | – | **B** | – | **B** | **B** |
| SafeKV [46] | **B** | **B** | **B** | – | – | **B** | **B** | U | **P** |
| KVGov [37] | **B** | **B** | **B** | – | – | **B** | **B** | U | **B** |
| CachePrune [7] | **B** | **B** | **B** | – | – | **B** | **B** | **B** | **B** |
| OTRO [32] | **B** | **B** | **B** | – | – | **B** | **B** | – | **B** |
| SparLeak [28] | **B** | **B** | **B** | **P** | U | **B** | **B** | **B** | U |
| Shape and Substance [29] | **B** | **B** | **P** | U | U | **B** | **P** | U | **P** |
| Bound | 19 | 16 | 16 | 4 | 2 | 16 | 15 | 3 | 11 |
| Applicable | 19 | 19 | 19 | 10 | 10 | 19 | 16 | 16 | 18 |

B: bound; P: partial; U: unreported in inspected scope; ?: unverified; –: inapplicable. Table 11 defines R1–R9. The crosswalk in Appendix F locates inference/cost witnesses.

Leak, Time Will Tell, KVGov and OTRO—no individual record carries both. The distinctions are concrete: prefix policy versus semantic processing; background-load mitigation versus QCP; a length instruction versus a padding estimate; partial flushing versus HMAC; and padded versus unpadded ORAM. SPEC, Dangol and CachePrune supply the five same-record inference/cost results listed in Table 13. Appendix F lists all eight families' exact supporting records.

This separation remains five of seven families after removing CachePrune and four of six after removing the final-search additions. Accepting partial cost and inference evidence reduces it to three of 12; a partial cost estimate can reconnect an operating point without supplying the fully bound cost. The strict majority is therefore specific to *explicit* bindings. It does not justify calling all remaining cost evidence a microbenchmark or an end-to-end measurement.

Positive combinations for deployment. The joint queries start from the 31 privacy/attribution records and exclude a record only when a required item is inapplicable. Table 13 names every positive result; S41 supplies query members. CachePrune's two target outcomes share one quality/cost package, while SparLeak provides intensity-specific inference/quality pairs. These examples support feasibility analysis at their measured configurations. The original R4/R5 ask specifically about fitting on protected data and a matched clean/protected fit. Their inapplicable status for a non-learning attacker does not answer whether another defense-aware attack is possible. A broader adaptive-attack question needs fresh source coding; treating those cells as unknown changes the denominators without inventing evidence (Table 14). Requiring explicit execution placement also removes SPEC's payload-cost record from the inference–cost query. The five retained cost objects in the original query are not interchangeable.

In a leave-one-family-out check, the two triple-binding records disappear only when CachePrune is removed; removing any of the other 18 families leaves both records. Removing all final-search additions leaves no triple-binding record among 15 eligible records. These are sensitivity descriptions for this corpus, not estimates of field-wide prevalence.

## 5.5. Execution evidence is distributed across measurement modes

Among the 31 privacy/attribution records, 13 are executed collections/prototypes, five trace transformations, five simulations/derived estimates, six component/calibration measurements and two unresolved placements. The 12-record network-receiver subset contains six executed, four transformed and two unresolved records. These modes are a mixture rather than a predominance of network replay. An executed prototype still supplies only its measured local scope.

Comparison permissions as a supporting index. The declared question groups generate 133 comparisons, of which 17 retain a descriptive comparison. Early Bird's prefix oracle and CachePrune's direct-recovery result can inform the same token-recovery shortlist while keeping their costs separate. Record splitting and question grouping determine this inventory; it guides a specific comparison rather than

TABLE 13. JOINT EVIDENCE IN THE 31 TYPE I/II MITIGATION RECORDS. REQUIRED INAPPLICABLE ITEMS EXCLUDE A RECORD FROM THAT QUERY. THE TWO TYPE III AHB RECORDS REMAIN IN THE HISTORICAL INDEX BUT ARE EXCLUDED HERE.

| Requested evidence | Bound record IDs | Interpretation at the source configuration |
|---|---|---|
| Inference + utility (3/20) | CACHEPRUNE-selective; CACHEPRUNE-context; SPARLEAK-replacement | Component quality; the two CachePrune targets share one package. |
| Inference + cost (5/21) | SPEC-padding-REST; DANGOL-capture; DANGOL-livebinding; both CACHEPRUNE records | Payload, transformed-capture, loopback and serving costs remain distinct. |
| Adaptation + inference (2/15) | UI-delay-offline; DANGOL-capture | Matched clean/protected fitting on a defended test. |
| Inference + utility + cost (2/19) | CACHEPRUNE-selective; CACHEPRUNE-context | One family and default configuration, two target outcomes. |
| All five items (0/13) | None in this extraction | No complete binding for this scoped query. |

SPEC reports a payload-size ratio at the evaluated configuration; its placement is only partially bound. This is not a measured wire-byte or end-to-end cost. Requiring explicit placement or a wire-cost object removes SPEC and leaves 4/21; the four retained records still use distinct cost objects.

TABLE 14. TWO FIT-APPLICABILITY AND COST-PLACEMENT INTERPRETATIONS OVER THE SAME 31 PRIVACY/ATTRIBUTION RECORDS. ENTRIES ARE BOUND/ELIGIBLE RECORDS. BROADENING MAPS FIT-INAPPLICABLE RECORDS TO UNKNOWN, NOT TO A VERIFIED ABSENCE OF ADAPTATION.

| Query | Fit scope | Broader unknown |
|---|---|---|
| Matched adaptation + inference | 2/15 | 2/22 |
| All five bindings | 0/13 | 0/19 |
| Inference + cost | 5/21 | 4/21† |

† Requires explicit R3 placement as well as R7/R9, or treats SPEC's payload-only cost as P for a wire-level query. The same four records remain, with distinct cost objects. Source cells and historical query labels remain unchanged.

measuring a field-wide defect rate. Complete decisions and coarser-granularity projections are retained in Appendix F. The reporting findings above are recomputed from cells and records, independently of pair counts.

**SoK Takeaway 2 (Bind function and cost to protection).** Inference reporting is widespread in the inspected mitigation sources; explicit utility and matched protected-data fitting contrasts are much less common. Check the measured operating point before combining protection and cost, and request a task outcome when a privacy intervention changes agent execution.

## 6. Which Evidence Supports Defense Transfer?

### 6.1. Separate the evaluated object before counting holdouts

RQ4 distinguishes three questions: does an *attack* recognize new inputs or systems; is a *defense condition* evaluated; and does a *fixed attacker or defense calibration* carry from a specified source to a target? Table 15 makes the distinction operational. Repeating a defense on several models can establish condition-specific outcomes without establishing a held-out model or transferring a fitted calibration.

The 82-family matrix in Appendix D is a discovery index for these questions. Its six axes are task/input/user, model, harness/topology, hardware, network/load and time. H marks reported separation, V condition variation/recalibration, X a scoped restriction, and ? unresolved extraction. The 13 task/input H entries include only four families in the historical 20-family mitigation inventory; all three model H entries concern attack/identification generalization outside that inventory. The two topology H entries occur in privacy/attribution mitigation families, but their held-out objects are workflow recognizers. The five model/topology H entries thus locate attack-generalization evidence, with no source-frozen defense-calibration transfer in those entries. The 13 task entries require their individual defended-condition linkage; they are not uniformly attack-only. S34 lists all H objects, while S38 retains their member identities. Varying or locally refitting a defense differs from transporting a source calibration.

TABLE 15. THREE EVIDENCE OBJECTS FOR RQ4. THE SAME FAMILY MAY SUPPLY SEVERAL OBJECTS IN SEPARATE EXPERIMENTS.

| Object | Required link | Permitted reuse |
|---|---|---|
| Attack generalization | Training/profiling source → held-out inputs or systems; attack outcome. | An attacker or recognition method covers the stated target domain. |
| Defended condition | Named mechanism, local fitting, defended test and outcome. | Retain that operating point; multi-model testing alone is not held-out transfer. |
| Defense transfer | Source-selected policy/calibration held fixed → target; local-calibration comparator and attacker fit specified. | Transport the evaluated policy only over the tested direction and scope. |

### 6.2. A changed generator and a tested defense are different bindings

LLM-Fin, ArchSnoop and LeakyLMs evaluate recognition of unseen model variants or architectures [79], [27], [50]. This supplies attack-side model generalization. In the traffic benchmark, graph-to-star and star-to-graph classification explicitly move the fitted attacker across topologies; the prompt/temperature countermeasures are separate evaluations [55, V-B–C]. Dangol's leave-composition-out test likewise examines recognition of a novel workflow shape,

separately from its protected-capture frontier [19, 9.8–9.12]. The reusable result in both topology entries is the recognizer's tested generalization.

There is nevertheless useful *defense-side variation* evidence. Time Will Tell applies the same fixed-length instruction to two models and reports different residual classification ASRs; this tests a common rule under changed generators [52, VII-B]. Whisper Leak evaluates mitigation conditions across provider-models, while its prompt holdout describes the classifier protocol [43, 3.2,6]. Preserve those per-condition results; a shared rule, a refitted local attack, a source-trained attack and a source-calibrated defense are four distinct comparisons.

Deployment decision. For the proposed change, name the source, target, held-out unit, quantities held fixed or refitted, and protected outcome. Carry forward only the corresponding binding. An attack-generalization result justifies including that attacker at the target; a defended-condition result supplies its local outcome; a source-to-target defense-calibration result supports only the stated transfer. This prevents both overclaiming portability and discarding useful local evidence. Unresolved hardware/time entries in the discovery matrix identify source follow-up, not absent experiments.

## 7. Synthesis: From Evidence to Reuse Decisions

The census (§5) covers all 39 records and the holdout matrix (§6) all 82 families. We now select three examples by the *comparison decision* they expose, not by effect size: a changed receiving path, cover with different state effects, and an adaptation/cost mismatch within a named mechanism. Their value is the decision they change: retain an aligned result, separate a wrongly joined package, or require a result at a changed execution boundary.

### 7.1. Case 1: bind the final target-bearing emitter

NetEcho's upstream packet-size shaping operates on a different projection from the tool-call sequence. Its plaintext-aware frontend then emits a new size trace (grouped token counts or character lengths), reopening a size channel after upstream protection [40]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [36, Sections 5.1–5.3]. Both change the receiver boundary and therefore require a separately bound inference result.

The data processing inequality applies when the downstream output receives only the protected view $Z$; a frontend that also receives content $X$ can re-emit a new signal $Y = f(Z, X)$ [80, Theorem 2.8.1]. NetEcho's source-backed re-emission example motivates capturing the final receiver view; the inequality alone is not a quantitative leakage estimate for that system.

### 7.2. Case 2: resolve the WRA versus AHB cover asymmetry

WRA discards decoy reads for Type I confidentiality; AHB's Type III evasion executes GUI events, a different objective and state contract [6], [17]. In our retail simulator, read-only and isolated-shadow cover preserve the reference database in 291/291 units, whereas a self-write inserted before the program changes it in 173/291 units (97 tasks, three selections; S9). These controlled units illustrate why discarding a response does not make an executed write inert. State equality is scoped to the simulator and does not establish production safety.

### 7.3. Case 3: bind adaptive retraining and cost objects

A matched adaptation contrast compares clean-trained $D_0$ and defense-trained $D_2$ on identical defended tests. Early Bird's 6ms/$4\%$ GPTCache semantic-processing cost does not price its $K$-prefix isolation; KVGov's $1.3\,\mu$s HMAC microcost omits lost reuse [3], [37]. These source-bound distinctions identify measurements needed for a deployment decision. Our small forward-citation pilot does not establish downstream misuse or its rate (S39). CachePrune instead binds its tested protection, component quality and serving cost [7]. Component quality and planner success remain distinct outcomes; the supplementary diagnostic records the uncertainty in a small live sample (S40).

### 7.4. A positive reuse path: from comparison to a candidate configuration

The Early Bird $K$-prefix record and CachePrune's direct-recovery record form one of the 17 descriptive matches: both address sensitive-token recovery through cache-reuse probes [3, V-A], [7, 5.1–5.6]. This permits an analyst to retain both mechanisms as candidates for that threat, while keeping the oracle and executed-serving measurements separate. The joint audit then changes the next step. CachePrune supplies a default configuration package linking recovery, generation quality and serving cost; the analyst can use that package for a source-scoped feasibility assessment. Early Bird's semantic-cache overhead cannot price its $K$-prefix candidate, so that candidate still needs a serving-cost measurement. Neither a pooled recovery ranking nor a new-hardware guarantee follows from this comparison.

What the three findings add. Record-level comparison turns a mechanism-level shortlist into scoped alternatives. Configuration-level binding distinguishes a reusable package from utility or cost measured elsewhere. Building on established WF and adaptive-attack practice [8], [9], [10], [11], the examples instantiate three LLM/agent joins: prefix versus semantic-cache primitives, discarded reads versus executed writes, and component quality versus planner outcomes. The examples identify specific measurements for a deployment; they do not establish that prior authors made a comparison error.

TABLE 16. CLAIM-CONDITIONED REUSE DECISIONS DERIVED FROM THE RECORD AND HOLDOUT AUDITS. EACH CHANGED BOUNDARY IDENTIFIES THE EVIDENCE THAT MUST BE BOUND BEFORE CARRYING A RESULT FORWARD.

| Reuse question | Evidence decision | Retain or measure next |
|---|---|---|
| Emitter / representation | An upstream or buffered capture can differ from the final receiving trace and delivery schedule. | Bind inference to the final receiver; retain the emitter's overflow, byte and delivery-delay scope (R2/R3/R9; §7). |
| Read cover → writes | Discarding a reply does not establish unchanged commits, authorization or workflow outcomes. | Bind the protected state and utility to the executed intervention; include cover calls, quotas and failures (R3/R8/R9). |
| Customer / model calibration | A reported split can separate inputs without changing the generator or calibration population. | State held-out units and calibration provenance; separate local fitting variability from source/domain changes; evaluate the requested transfer axis (R5/R6; §6). |
| Attacker observation | An outcome for one projection or fitted attacker does not supply a result for a richer receiving view. | Evaluate comparable defended tests under the intended observation and adaptation contract; retain receiver-specific inference metrics (R2/R5/R7). |
| Stored trace → online path | A trace transformation or simulated span does not measure executed queues, completion or online latency. | Bind the receiver, harness and resource denominator to the online condition; establish the relevant topology/network holdout (R3/R6/R9). |
| Bound cache-reuse package | Early Bird prefix recovery and CachePrune direct recovery permit a scoped candidate comparison; CachePrune binds a shared utility/cost package. | Retain that package for its measured configuration; measure prefix-policy serving cost separately, and re-evaluate any changed workload/hardware. |
| Component → workflow utility | Component quality or logical state conformance does not determine a planner's response to scheduling and deadlines. | Report utility at the requested workflow boundary and the same intervention/configuration as protection and cost (R3/R8/R9). |

## 7.5. A deployment query determines the evidence to retain

Start with the intended receiver, protected outcome and intervention. Record-level comparison identifies whether the cited result measures that object. Configuration-level joint binding identifies which inference, utility and cost outcomes share its configuration. Deployment decisions use the holdout matrix to identify evidence for the proposed change and specify the remaining measurement. Table 16 makes these outputs operational: a new emitter needs a receiver-specific capture, writes need state/authorization checks, source-calibration comparisons need a local fitting-variability control, and an online scheduler needs its own planner utility and latency measurement. The calibration control follows from the comparison design: changing the source can also change the finite training sample used to fit the defense.

The following worked applications show how the procedure represents external reevaluations and internal examples. Appendix E retains the detailed controlled case study.

## 7.6. External worked applications: retain, separate, remeasure

We apply the procedure to a fixed external frame: both defenses in Deep Fingerprinting, all three in WFDefProxy, and all nine in Mathews et al.'s reevaluation, with DynaFlow's live implementation kept as a separate record [81], [82], [9]. The resulting 15 measurement records represent 14 defense–study pairs in three studies. The ledger retains favorable outcomes alongside weaknesses, fixes the Standard setting for the nine-defense table, and records a locator and a concrete measurement decision for every row (S35).

The worked output separates decisions that a mechanism name conflates (Table 17). Stronger protected-data fitting requires revisiting WTF-PAD's attack result, while Walkie-Talkie's symmetric-pairing result remains a valid scoped outcome. FRONT's live protection improves even as its bandwidth cost changes. A record-level audit therefore preserves the security observation and replaces the cost estimate separately. DynaFlow's live result needs a separate execution/workload record, preventing a simulated cost from being attached to a live attack result.

The complete frame produces 11 protected-fit/view measurement requirements and four execution/cost requirements. Four favorable controls remain usable in their measured scopes: symmetric Walkie-Talkie, live FRONT, live Tamaraw and one-split TrafficSliver. The actionable output is a scoped retain, separate or remeasure decision for each record. The examples faithfully encode established WF practice. Outcomes were visible during extraction, so agreement is an illustration of the representation rather than evidence of prediction accuracy or an improvement over ordinary expert reading. These three WF studies do not establish LLM/agent effectiveness; the cache and tool examples provide the domain-specific application.

## 7.7. Seven internal illustrative judgments

The existing controlled study supplies illustrative applications: seven written judgments specify which apparent protection claims should survive a changed target, receiver or attacker. The protocol-only repository state predates the first archived result commit on 21 October 2026; S35 preserves the ordering receipt, original labels and current reassessment. Table 18 brings all seven into the main paper.

TABLE 17. EXTERNAL APPLICATIONS: RETAIN A SCOPED RESULT, SEPARATE AN OUTCOME PACKAGE, OR REQUEST A SPECIFIC MEASUREMENT. FULL 15-RECORD LEDGER: S35. ACCURACY AND TPR RETAIN THEIR ORIGINAL POPULATIONS.

| Published contrast | Evidence-reuse decision |
|---|---|
| WTF-PAD [81] | Refit: DF accuracy 90.7%; prior attackers in the same table reach at most 69.0%. |
| Walkie-Talkie [81] | Retain symmetric-pairing scope: DF accuracy 49.7%; do not inherit it for asymmetric pairing. |
| FRONT [82] | Separate outcomes: live DF TPR improves, 72.06% to 42.79%, while data overhead rises, 52% to 75%. |
| Tamaraw [82] | Retain low live TPR, 6.43%; use live bandwidth overhead 179%, rather than simulated 207/224%. |
| DynaFlow [9] | Separate live record: 379% mean per-site time overhead; changed URLs/crawler prevent a matched simulation–live effect. |

TABLE 18. CURRENT OUTCOMES OF ALL SEVEN TIME-ORDERED INTERNAL JUDGMENTS. OBSERVED REFERS TO THE SPECIFIED FINITE EXAMPLE, NOT A FRAMEWORK-ACCURACY ESTIMATE.

| ID | Reuse claim tested | Finding and next measurement |
|---|---|---|
| P1 | Buffered protection/cost carries to a stream | Observed: verbosity accuracy 100%; capture the stream emitter. |
| P2 | Low clean-trained accuracy establishes protection | Observed: $D_0$ 50%, $D_2$ 100%; refit on protected data. |
| P3 | Padded events hide response identity | Observed: top-1 58–62%, chance 2.1%; test the whole trace. |
| P4 | Equal target-verb count hides cancellation | Observed: count BA 50%, suite BA 76.9%; expand the view. |
| P5 | Protected dispatch implies a protected tool link | Source premise unmet; excluded from favorable evidence. Establish dispatch protection first. |
| P6 | Framing alone hides intent | Partial: bucketed return leaks; resolve the remaining target cells. |
| P7 | Joint cover/framing protects both targets | Not established: retain the joint screen before accepting protection. |

Four judgments have observations in finite examples; one has a partial observation. P5 has a failed source premise and is excluded from favorable evidence for either transfer failure or framework success. P7's proposed joint protection remains unestablished. The examples specify the receiving view and protected-data fit used in each calculation. Small samples and mutable model aliases limit their interpretation; the case study is illustrative rather than validation of the framework. Repository ordering establishes chronology only; author-designed examples do not measure the accuracy of the procedure. The decisions differ by failure mode: P1–P4 require the changed emitter, attacker or trace view to be measured; P5 requires establishing source protection first; P6/P7 require evidence for the still-open claim. Appendix E retains the measurement scopes and statistical qualifications.

TABLE 19. REUSABLE REPORTING CARD. FILL EACH EVIDENCE CELL WITH A VERSION AND LOCATOR OR "UNKNOWN"; THE FINAL ACTION IS CONDITIONAL ON THE PROPOSED DEPLOYMENT.

| Field | What to enter |
|---|---|
| Receiver/target | Hidden property, accessible signal and permissions. |
| Intervention | Primitive, execution mode, state and trust boundary. |
| Attacker/test | Training or calibration condition; defended test population. |
| Function | Component quality or task outcome; shared configuration. |
| Cost | Resource, units, denominator, baseline and included operations. |
| Extension | Changed workload/system axis; available holdout; retain, separate or measure next. |

# 8. Implications and Ethics

## 8.1. Deployment and reporting implications

The decision table distinguishes receiving permissions from intervention capabilities. An access-network operator, VPC monitor and receiving tool service expose different views; remote paths can reveal destinations even with Encrypted ClientHello [83]. A result for one ingress projection therefore needs a separately bound claim before it is applied to host instrumentation or downstream commit logs.

Likewise, read queries, provider-isolated execution and gateway-routed cover have different state contracts. Kubernetes dry-run, Stripe sandboxes and transactional rollback offer isolation mechanisms with their own limits [84], [85], [86]; idempotence alone is insufficient, and sequence changes can survive rollback [87]. Evidence reuse requires the trusted boundary, authorization policy, resource quotas and protected outcome to accompany the mechanism name. The corpus's component outcomes remain useful when their execution and cost scopes match the intended query.

A usable reporting card. Table 19 is a reusable six-field template. Its last column asks for a deployment decision after recording the source evidence. Table 20 fills it for two venue-linked sources and the running cache example, preserving their different scopes. The editable artifact template includes an explicit unknown state and source-locator slot for every field.

## 8.2. Ethics and artifact scope

The audit uses source-linked public research records. Stakeholders include tenants, shared-capacity providers and users affected by observable behavior or state changes. Bot-detection evasion and research-intent privacy have different authorization relations [17], [6]. The appendix case study uses cloned public benchmark databases and controlled sockets; model interactions use public tasks and simulated users. No production account is modified.

AI tools assisted candidate-field extraction, implementation, source reconciliation and prose revision. The author-review coverage is stated in §2. Reliability controls retain

TABLE 20. FILLED REPORTING CARDS FROM SOURCE RESULTS. EARLY BIRD IS VENUE-LINKED (INSPECTED AUTHOR VERSION); TDXRAY IS AN S&P PAPER; CACHEPRUNE IS A CITED PREPRINT AND SERVES AS THE RUNNING CONFIGURATION EXAMPLE. NO NEW BENCHMARK IS IMPLIED.

| Field | Early Bird [3, V-A–B] | TDXRay [30, 6.1, Fig. 8, Table 3] | CachePrune [7, 5.1–5.6] |
|---|---|---|---|
| Receiver/target | Timing oracle; sensitive prefix recovery. | Memory-access view; oblivious lookup defense component. | Cache-reuse probes; direct and contextual token recovery. |
| Intervention | Prefix-granularity oracle; semantic-cache prototype separately. | Oblivious mapping microbenchmark. | Selective cache sharing at default configuration. |
| Attacker/test | Granularity-specific calibration; matched adaptation unresolved. | Component cost, no bound protected end-to-end inference row. | Recovery evaluation; default quality/cost package; matched training-fit contrast inapplicable; broader adaptive attacks unassessed. |
| Function | No bound prefix-policy utility. | No bound full-inference utility. | QA/summarization quality at stated configuration. |
| Cost | Semantic-cache overhead belongs to a different primitive. | Lookup microcost; not complete serving cost. | Serving throughput/cost; preserve default package. |
| Extension/action | Retain prefix candidate; measure its own serving cost. | Retain component cost; measure protected service and its function. | Retain scoped package; test changed workload/hardware. |

source versions and locators, separate unavailable from unreported evidence, preserve corrections, and recompute aggregation and joins from the record cells before compilation. These controls detect inconsistent accounting and support source inspection; they do not estimate an AI extraction-error rate. Authors retain responsibility for source interpretation. Case-study observations and protocols remain separate from the literature atlas.

## 9. Conclusion

In the inspected privacy/attribution mitigation sources, inference outcomes are reported more consistently than utility or a matched clean/protected fitting contrast. This pattern survives accepting partial bindings and restricting outcomes to a common family population. Agent privacy/attribution studies provide partial usefulness evidence but no explicit task-utility binding in the four-family subset. Five of eight families with explicit inference and cost place the two in different records; the operating point determines whether they can be combined.

The SoK connects these reporting patterns to deployment decisions through scoped result records. Positive packages such as CachePrune and SparLeak show what can be retained, while the holdout synthesis identifies whether a changed system has attack-side or defense-side evidence. The released record identities, sensitivity checks and filled reporting cards make the findings inspectable and turn the remaining bindings into concrete measurement requirements.

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## Appendix A. Audit Trail

The artifact preserves the 82-family map, 39-record ledger, 180 family cells and 351 record cells with source versions and locators. S33 publishes query membership and sensitivity checks. Case-study protocols, observations and uncertainty calculations retain their original cohorts and provenance.

## Appendix B. Result-record index

Table 21 indexes all 39 records by measurement mode and source locator. Marks distinguish six contextual controls (‡) and six final-search additions (†); all enter the pair inventory. The ledger retains full locators, versions and hashes.

## Appendix C. Case-Study Artifact Index

Table 22 maps the single appendix case to historical artifact keys and supplementary methods. These keys preserve original task/customer memberships; S30 retains the corrected permutation/Holm tests and calibration resampling.

## Appendix D. Family-Level Holdout Evidence

This is a family-level discovery index, not a count of defense-transfer tests. The three model H entries concern attack/identification generalization; even an H in a mitigation family must be linked to its defended condition before reuse. The main RQ4 table separates those evidence objects, and S34 identifies every H entry. Table 23 covers 82 families and 492 cells. Axes: T task/input/user; M model; C harness/topology; H hardware; N network/load; D time. Codes: H reported holdout; V condition comparison/recalibration; X scoped restriction; ? unresolved; – conceptual/inapplicable. Scope C/A denotes core/adjacent. H records separation, not transfer success; ? does not establish an absent experiment. S28 retains source locators and qualified interpretations.

## Appendix E. Illustrative Case Study and Reconstruction Scope

This appendix explains the role of the archived implementation examples. They instantiate the same distinctions used in the literature atlas: final receiver versus upstream trace, executed state versus discarded responses, and online planner utility versus reused trajectories. They are illustrative applications, not validation data for the comparison rules. The complete numerical grids, negative outcomes, intervals and corrected tests remain unchanged in S11–S32.

The transport example has 48 saved responses on 24 topics (12 train/12 test). The public *retail* setting has 115 tasks from 53 customers; its frozen split contains 46 test tasks from 24 customers, including 5 cancellation-positive and 7 return-positive customers. The separate live planner example has 100 episodes: 37/50 baseline and 33/50 defended successes, a difference of $-8.0$ points with customer interval $[-23.9, 6.5]$. All eight calibration contrasts have Holm-adjusted $p = 1.000$; the original absolute protection screen remains unmet. Neither these unresolved contrasts nor resampling the same customers validates the framework.

The artifact separates *analysis reconstruction* from *new model collection*. Archived requests, responses, traces, calibration hashes and predictions permit the reported calculations to be reconstructed. Fresh calls to mutable Sol/Luna/MiniMax/GLM aliases, with documented unequal budgets, cannot reproduce an identical generator. No fixed-weight replication claim is made. Historical protocols and

model identities remain explicit in S14–S32; the literature synthesis does not depend on obtaining a favorable result from these examples.

## Appendix F. Reconstructing the Audit in This PDF

Table 24 gives the reporting-pattern sensitivity to removing CachePrune and final-search additions. The latter removes six records from four families (the † marker in the record index); six contextual ‡ controls are excluded throughout. Removing both CachePrune and all final-search additions equals the latter row. This preserves the inference–utility and matched-adaptation patterns, and the agent subset remains 0/4 B or 2/4 B+P. S38 gives exact identities and the common-denominator calculation.

Table 25 retains the historical 33-record extended inventory, including two Type III AHB records. Excluding those two reproduces the 31-record privacy/attribution query in Table 13; S41 supplies the filtered members. Table 26 explains every family inference/cost conjunction. The 39 record IDs, measurement modes and source locators are in Appendix B.

Pair inventory and granularity. Table 27 covers all 133 within-question pair decisions. The 105 different-object pairs and eight of 15 within-family pairs reflect the declared split and grouping rules; a normal difference in research scope is not a source-paper defect. Table 28 projects the unchanged judgments onto coarser units. Mixed edges retain all constituent statuses, and within-unit edges disappear from the index without being resolved. Family-only indexing yields 44 edges, eight mixed, and hides all 15 within-family pairs. The pair CSV preserves the locators and individual reasons. These indexing sensitivities support audit reconstruction; the reporting findings use the record cells directly.

## Appendix G. Mapped Privacy Families without a Linked Mitigation Record

Within the frozen corpus, the union of Type I and II contains 52 families; 19 have a scoped mitigation record. The remaining 33 families are enumerated below so that this coverage statement can be audited against the core map. This is a statement about the record-selection rules and inspected sources: several have contextual guardrails, noise, filters, decoy boundaries, or unbound claims. It does not assert that none of their papers or subsequent work evaluates any defense.

The Type I families without a linked qualifying mitigation record are Edge EM leakage, MoEcho, Behind Bars, PAGE404, Spill the Beans, SparSEEty, Detokenization Leaks, LLM-WFIN, SplitDecoy, Prompt Echo, remote keylogging, cache contention, routing metadata, ML privacy channels, prompt-cache audit, InputSnatch, Leaky Language Models, PromptPeek, MoE prompt stealing, AgentPrint, SIGLEAK, PersonaFingerprint, and AgentTell [20], [22], [24], [25], [26], [31], [33], [34], [36], [41], [1], [44], [45], [47], [48], [49], [50], [2], [53], [4], [59], [64], [66].

The Type II families without a linked qualifying mitigation record are Energon, ArchSnoop, LLMs Have Rhythm, Web-agent unmasking, AgentPrint, ProcGrep, AgentProv, Lidar, Campaign attribution, Whose Agent?, and FP-Agent [21], [27], [38], [56], [4], [60], [61], [62], [63], [65], [67]. AgentPrint belongs to both lists and is counted once in the union.

TABLE 21. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

| Record ID / family | Mode | Boundary; source locator |
|---|---|---|
| UI-delay-offline [5] | Tr | Offline delay; no live task outcome. 6.1 F5; A.5 |
| WRA-decoy-execution [6] | Ex | Executed decoys; usefulness proxy only. 8.1–8.3 T8–11 |
| NET-upstream-local [40] | Ex | Protected deployment/local serving; topology scoped. IV-C–E TIII–IV |
| NET-dummy-padding-simulation [40] | Tr | Synthetic insert/pad; no live execution. VII TIX |
| EARLY-K-calibration-oracle [3] | Si | $K$-specific fit then oracle; not live KV trial. V-A TVI; IV-A |
| EARLY-semantic-deidentification [3] | Ex | GPTCache prototype cost; not KV-granularity cost. V-B |
| WHISPER-randompadding-capture [43] | Ex | New padded captures; rounded medians. 6.1 T3–4 |
| WHISPER-batching-replay [43] | Tr | Offline batching; model-dependent result. 6.2 F7 |
| WHISPER-injection-replay [43] | Tr | Synthetic packets; no live zero-delay claim. 6.3 T5 |
| SPLICE-CTBF-signal [51] | Ex | Prototype background load; tested signal only. VII-C TII |
| SPLICE-QCP-cost [51] | Co | QCP component cost; excludes combined CTBF. VII-C TII |
| TIME-length-instruction [52] | Ex | Length instruction; unequal outputs remain. 7.2 |
| TIME-padding-estimate [52] | Si | Analytic plaintext cost; not network bytes. 7.3 |
| SPEC-padding-REST [42] | Un | Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2 |
| DANGOL-capture [19] | Tr | Capture transform; distinct from live population. 9.8–9.9 T6 |
| DANGOL-livebinding [19] | Ex | Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5 |
| SIG-common-guardrail‡ [59] | Ex | Common instruction; not a causal guardrail effect. 4.1–4.2; App. C |
| SAFE-post-trigger [46] | Ex | Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4 |
| AHB-online-executed [17] | Ex | Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3 |
| AHB-offline [17] | Tr | Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3 |
| KVGOV-partialflush-sim [37] | Si | Deterministic simulation; not live protected recovery. 5.1–5.3 |
| KVGOV-hardware-gap [37] | Co | Hardware gap calibration; not protected ASR. 5.7–5.8 |
| KVGOV-HMAC-cost [37] | Co | HMAC microcost; not cache reuse/task loss. 5.9 |
| TRAFFIC-prompt-reformulation [55] | Ex | Collected prompt rewrites; task utility unmeasured. V-C1 F9a |
| TRAFFIC-temperature [55] | Ex | Collected decoding changes; no monotone benefit. V-C1 F9a |
| CARLINI-fixed-rate [39] | Un | Rate/replayed cost; per-token, not request delay. VII F11–12 |
| GAO-zero-copy-off [23] | Co | Loader component costs; not end-to-end TTFT. 7.1 |
| TDXRAY-oblivious-map [30] | Co | Lookup microbenchmark; not full inference cost. 6.1 F8 T3 |
| AGENTTELL-fixed-instruction‡ [66] | Ex | Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1 |
| FP-CLOUDFLARE‡ [67] | Ex | Live access control; not user confidentiality. 6.4 T5 |
| BAR-co-running-noise‡ [24] | Ex | MIG noise test; not an LLM defense trial. 7.2–7.3 T8 |
| RHYTHM-VPN‡ [38] | Ex | Live VPN shift; baseline-trained attacker only. VI-B; VII |
| DEB-filter-origin‡ [47] | Ex | Filter creates attack view; not mitigation of it. Threat model; 4.2 |
| CACHEPRUNE-selective† [7] | Ex | Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A |
| CACHEPRUNE-context† [7] | Ex | Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A |
| OTRO-unpadded† [32] | Co | ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII |
| OTRO-epochpad† [32] | Si | Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII |
| SPARLEAK-replacement† [28] | Ex | Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4 |
| SHAPE-padding-estimate† [29] | Si | Constant-work proposal/estimate; no executed pair. IX-B1 |

TABLE 22. ARTIFACT INDEX FOR THE APPENDIX CASE STUDY. COMPONENT NAMES DESCRIBE THE ARCHIVED MEASUREMENT BOUNDARY; THE RIGHT COLUMN LOCATES IMPLEMENTATION AND FULL-GRID DETAIL.

| Case-study component | Artifact keys | Observation/execution boundary and comparison | Detailed source |
|---|---|---|---|
| Transport emission | E1, E3, E5 | Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons. | S9/S11; S12 |
| State and link | E2, E6, E7 | Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views. | S9/S11; S12 |
| Trajectories and localization | E9, E10, E12 | Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries. | S14; S15/S21 |
| Timing and WF controls | E13, E14 | Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons. | S19; S20 |
| Extensions | E8 | Stand-in-planner pipeline check. | S13 |
| | E11 | Retrospective training-fold operating-point selection with the historical suite. | S15 |
| | E15 | Additional same-provider Luna cohort. | S22 |
| | E16 | Same-operation semantic-attribute diagnostic on reused Sol data. | S23.6 |
| | R11/R12 | Two served model endpoints: native trajectories, frozen training calibration and TLS write localization. | S24 |
| | R14 | Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the endpoint suite. | S25 |
| | R15 | Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines. | S26 |

TABLE 23. COMPLETE HOLDOUT-CODE MATRIX; CODES AND AXES ARE DEFINED IN APPENDIX D. SOURCE-QUALIFIED ROW NOTES ARE IN SUPPLEMENT S28.

| Family | Scope | T | M | C | H | N | D | Family | Scope | T | M | C | H | N | D |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| F01 AgentPrint [4] | C | ? | ? | ? | ? | ? | ? | F42 ML privacy channels [47] | C | ? | V | ? | ? | ? | ? |
| F02 UI Traces [5] | C | V | V | V | ? | ? | ? | F43 MoE stealing [53] | C | ? | ? | ? | ? | ? | ? |
| F03 WRA [6] | C | H | ? | ? | ? | ? | ? | F44 Energon [21] | C | ? | V | ? | ? | ? | ? |
| F04 FP-Agent [67] | C | H | ? | X | ? | ? | ? | F45 SafeKV [46] | C | ? | V | ? | ? | ? | ? |
| F05 Prompt length [1] | C | H | ? | V | ? | V | V | F46 InferNet [88] | A | ? | V | ? | V | ? | ? |
| F06 PromptPeek [2] | C | V | ? | ? | ? | V | ? | F47 ArchSnoop [27] | C | ? | H | ? | V | ? | ? |
| F07 NetEcho [40] | C | V | ? | V | ? | ? | ? | F48 Coding fingerprints [12] | A | X | ? | X | ? | ? | X |
| F08 Early Bird [3] | C | ? | V | ? | ? | ? | ? | F49 AHB [17] | C | ? | ? | X | ? | ? | ? |
| F09 InputSnatch [49] | C | H | ? | ? | ? | ? | ? | F50 WildAuth [13] | A | H | ? | X | ? | ? | ? |
| F10 Whisper Leak [43] | C | H | V | ? | ? | ? | ? | F51 Cycles [58] | C | X | ? | X | ? | ? | X |
| F11 Spill the Beans [26] | C | V | ? | ? | ? | ? | ? | F52 LumiMAS [57] | C | ? | V | V | ? | ? | ? |
| F12 SpliceLeak [51] | C | V | ? | ? | ? | ? | ? | F53 SymDiag [89] | A | V | V | ? | ? | ? | ? |
| F13 Time Will Tell [52] | C | ? | ? | ? | ? | V | V | F54 LeakyLMs [50] | C | ? | H | ? | V | ? | ? |
| F14 Prompt Echo [41] | C | H | V | ? | ? | ? | ? | F55 KVGov [37] | C | ? | ? | ? | V | ? | ? |
| F15 I Know What You Said [23] | C | H | V | V | V | ? | ? | F56 Data-centric survey [90] | A | - | - | - | - | - | - |
| F16 Speculation [42] | C | H | ? | ? | ? | V | ? | F57 Serving RCA [35] | C | ? | ? | X | ? | ? | ? |
| F17 Behind Bars [24] | C | ? | V | ? | V | ? | ? | F58 ZeroPrint [14] | A | ? | V | ? | ? | ? | ? |
| F18 Prompt caching [48] | C | ? | V | ? | ? | ? | V | F59 Endpoint fingerprints [15] | A | ? | ? | V | ? | ? | V |
| F19 Whose Agent [65] | C | ? | ? | ? | ? | ? | ? | F60 StepFinder [16] | A | H | ? | ? | ? | ? | ? |
| F20 Workflow traffic [19] | C | ? | ? | H | ? | ? | ? | F61 Web-agent detection [56] | C | ? | V | ? | ? | ? | ? |
| F21 GPU contention [44] | C | ? | ? | V | ? | V | ? | F62 Cross-campaign [63] | C | V | ? | V | ? | ? | ? |
| F22 AgentTell [66] | C | V | ? | ? | ? | V | ? | F63 AgentProv [61] | C | ? | V | ? | ? | ? | ? |
| F23 Traffic benchmark [55] | C | ? | ? | H | ? | ? | ? | F64 LiDAR [62] | C | ? | V | V | ? | ? | ? |
| F24 Remote timing [39] | C | ? | V | ? | ? | ? | V | F65 LLM-WFIN [34] | C | V | ? | ? | ? | ? | ? |
| F25 Persona fingerprint [64] | C | ? | ? | X | ? | ? | ? | F66 MIRAGE [54] | C | V | V | ? | ? | ? | ? |
| F26 TDXRay [30] | C | ? | ? | ? | ? | ? | ? | F67 Routing metadata [45] | C | V | ? | V | ? | ? | ? |
| F27 Detokenization [33] | C | ? | ? | X | ? | ? | ? | F68 Shape and Substance [29] | C | V | V | V | V | ? | ? |
| F28 Spillage [91] | A | V | ? | ? | ? | ? | ? | F69 SplitDecoy [36] | C | ? | ? | ? | ? | ? | ? |
| F29 PlanTwin [92] | A | V | ? | ? | ? | ? | ? | F70 PAGE404 [25] | C | V | V | ? | ? | ? | ? |
| F30 SICC [93] | A | ? | ? | ? | ? | ? | ? | F71 Edge EM [20] | C | H | ? | ? | ? | ? | ? |
| F31 TOPR [94] | A | V | ? | ? | ? | ? | ? | F72 SparLeak [28] | C | V | V | V | ? | ? | ? |
| F32 ClawGuard [18] | C | ? | ? | V | ? | ? | V | F73 Sparseety [31] | C | V | V | V | ? | ? | ? |
| F33 SIGLEAK [59] | C | V | ? | ? | ? | ? | ? | F74 OTRO [32] | C | ? | ? | ? | ? | ? | ? |
| F34 ProcGrep [60] | C | H | ? | V | ? | ? | ? | F75 CachePrune [7] | C | V | V | ? | ? | ? | ? |
| F35 SecEmb [95] | A | ? | X | ? | ? | ? | ? | F76 Privacy foundations [70] | A | - | - | - | - | - | - |
| F36 Mobile agents [96] | A | ? | ? | ? | ? | ? | ? | F77 DP-MemView [97] | A | ? | ? | ? | ? | ? | ? |
| F37 Hidden risks [98] | A | ? | ? | ? | ? | ? | ? | F78 PriCon [99] | A | ? | ? | ? | ? | ? | ? |
| F38 Browser privacy [100] | A | ? | ? | V | ? | ? | ? | F79 SCAAR [101] | A | ? | ? | ? | ? | ? | ? |
| F39 LLMs Have Rhythm [38] | C | ? | V | ? | ? | H | V | F80 CacheSolidarity [102] | A | ? | ? | ? | ? | ? | ? |
| F40 MoEcho [22] | C | H | V | ? | ? | V | ? | F81 Vision encoder [103] | A | ? | ? | ? | ? | ? | ? |
| F41 LLM-Fin [79] | A | ? | H | ? | ? | ? | ? | F82 Split gradients [104] | A | ? | ? | ? | ? | ? | ? |

TABLE 24. SENSITIVITY OF REPORTING PATTERNS. FRACTIONS GIVE EXPLICIT / APPLICABLE FAMILY COUNTS; BRACKETED NUMERATORS ALSO ACCEPT P. REMOVING FINAL-SEARCH ADDITIONS EXCLUDES SIX RECORDS FROM FOUR FAMILIES; CONTEXTUAL CONTROLS ARE EXCLUDED IN EVERY ROW. IC SEPARATE MEANS NO SINGLE RECORD CARRIES BOTH EXPLICIT INFERENCE AND COST.

| Inventory | R7 | R9 | R8 | R4 | R5 | Agent R8 | IC separate |
|---|---|---|---|---|---|---|---|
| All mitigation | 16/17 [17] | 11/19 [15] | 4/17 [6] | 4/11 [9] | 2/11 [2] | 0/4 [2] | 5/8 |
| Without CachePrune | 15/16 [16] | 10/18 [14] | 3/16 [5] | 4/11 [9] | 2/11 [2] | 0/4 [2] | 5/7 |
| Without final-search records | 13/13 [13] | 9/15 [12] | 2/14 [4] | 4/9 [8] | 2/9 [2] | 0/4 [2] | 4/6 |

TABLE 25. ALL 33 MITIGATION RECORDS: QUERY MEMBERSHIP. B: ELIGIBLE AND BOUND; E: ELIGIBLE WITH AT LEAST ONE INCOMPLETE BINDING; X: REQUIRED ITEM INAPPLICABLE. IU=INFERENCE/UTILITY; IC=INFERENCE/COST; AI=ADAPTATION/INFERENCE; IUC=TRIPLE; ALL=R4/R5/R7/R8/R9. FULL R1–R9 SOURCE RATIONALES REMAIN IN S28.

| Record / cited family | IU | IC | AI | IUC | All |
|---|---|---|---|---|---|
| UI-delay-offline [5] | E | x | B | x | x |
| WRA-decoy-execution [6] | E | E | x | E | x |
| NET-upstream-local [40] | E | E | E | E | E |
| NET-dummy-padding-simulation [40] | E | E | E | E | E |
| EARLY-K-calibration-oracle [3] | E | E | E | E | E |
| EARLY-semantic-deidentification [3] | x | x | x | x | x |
| WHISPER-randompadding-capture [43] | E | E | E | E | E |
| WHISPER-batching-replay [43] | E | E | E | E | E |
| WHISPER-injection-replay [43] | E | E | E | E | E |
| SPLICE-CTBF-signal [51] | E | E | x | E | x |
| SPLICE-QCP-cost [51] | x | x | x | x | x |
| TIME-length-instruction [52] | E | E | E | E | E |
| TIME-padding-estimate [52] | x | x | x | x | x |
| SPEC-padding-REST [42] | x | B | E | x | x |
| DANGOL-capture [19] | E | B | B | E | E |
| DANGOL-livebinding [19] | E | B | E | E | E |
| SAFE-post-trigger [46] | E | E | x | E | x |
| AHB-online-executed [17] | B | E | E | E | E |
| AHB-offline [17] | E | x | E | x | x |
| KVGOV-partialflush-sim [37] | E | E | x | E | x |
| KVGOV-hardware-gap [37] | x | x | x | x | x |
| KVGOV-HMAC-cost [37] | x | x | x | x | x |
| TRAFFIC-prompt-reformulation [55] | E | E | E | E | E |
| TRAFFIC-temperature [55] | E | E | E | E | E |
| CARLINI-fixed-rate [39] | x | x | x | x | x |
| GAO-zero-copy-off [23] | x | x | x | x | x |
| TDXRAY-oblivious-map [30] | x | x | x | x | x |
| CACHEPRUNE-selective [7] | B | B | x | B | x |
| CACHEPRUNE-context [7] | B | B | x | B | x |
| OTRO-unpadded [32] | x | x | x | x | x |
| OTRO-epochpad [32] | x | E | x | x | x |
| SPARLEAK-replacement [28] | B | E | E | E | E |
| SHAPE-padding-estimate [29] | E | E | E | E | E |

TABLE 26. EVERY FAMILY WITH B IN BOTH INFERENCE AND COST COLUMNS OF TABLE 12. SEPARATE COLUMN WITNESSES DO NOT AUTOMATICALLY FORM ONE ROW-LEVEL RESULT. IDS EXPAND IN APPENDIX B.

| Family | Inference record(s) | Cost record(s) | Same-record inference + cost |
|---|---|---|---|
| [3] | EARLY-K-calibration-oracle | EARLY-semantic-deidentification | – |
| [51] | SPLICE-CTBF-signal | SPLICE-QCP-cost | – |
| [52] | TIME-length-instruction | TIME-padding-estimate | – |
| [42] | SPEC-padding-REST | SPEC-padding-REST | SPEC-padding-REST |
| [19] | DANGOL-capture; DANGOL-livebinding | DANGOL-capture; DANGOL-livebinding | DANGOL-capture; DANGOL-livebinding |
| [37] | KVGOV-partialflush-sim | KVGOV-HMAC-cost | – |
| [7] | CACHEPRUNE-selective; CACHEPRUNE-context | CACHEPRUNE-selective; CACHEPRUNE-context | CACHEPRUNE-selective; CACHEPRUNE-context |
| [32] | OTRO-epochpad | OTRO-unpadded | – |

TABLE 27. ALL 133 WITHIN-QUESTION DECISIONS; 608 OTHER PAIRS HAVE NO DECLARED COMPARISON. COUNTS DESCRIBE THIS INVENTORY AND ITS GROUPING RULES.

| Question | Match | Different object | Unresolved |
|---|---|---|---|
| behavioral humanization | 1 | 2 | 0 |
| cache service | 2 | 37 | 6 |
| fixed instruction | 0 | 0 | 1 |
| generator fingerprint | 3 | 0 | 0 |
| local memory | 0 | 14 | 1 |
| traffic envelope | 11 | 52 | 3 |
| Total | 17 | 105 | 11 |

TABLE 28. SENSITIVITY TO THE COMPARISON UNIT. C/M/U: UNANIMOUS DESCRIPTIVE MATCH/OBJECT MISMATCH/UNRESOLVED; MIX: CONSTITUENT JUDGMENTS DIFFER. HIDDEN: ORIGINAL WITHIN-QUESTION PAIRS ABSORBED INSIDE A UNIT. NO NEW PAIR IS ADJUDICATED.

| Unit | Units | C | M | U | Mix | Hidden |
|---|---|---|---|---|---|---|
| Scoped result records | 39 | 17 | 105 | 11 | 0 | 0 |
| Shared configuration package | 38 | 16 | 98 | 9 | 1 | 1 |
| Family and measurement mode | 35 | 10 | 83 | 9 | 1 | 4 |
| Family only | 26 | 2 | 31 | 3 | 8 | 15 |